AI ACADEMY · PYTHON READINESS BRIDGE · STUDENT LESSON

Reading structured data

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Python readiness portfolio

THE CONNECTED WEEK

Read the fields. State the row policy. Report the coverage.

Aggregate structured records without disguising missing observations as zero.

Keep one habit: Required cupboard loop → observed-count graphic → complete coverage program → six policy cases → selected score report.

MAIN LESSON · SECTION 1

Which rows contributed, and which facts are still missing?

> Keep observed totals, mean denominators and coverage separate.

Your learning goals

I can **process** a list of records consistently.

I can compute a total from named **field**s.

I can preserve record meaning while tracing a loop.

Recall before notes

- Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.
- Without opening the old lesson, explain one key idea from Dictionaries and records. Give an example and a mistake that the idea helps you avoid.
- Without opening the old lesson, explain one key idea from Functions and arguments. Give an example and a mistake that the idea helps you avoid.
- Without opening the old lesson, explain one key idea from Branching and boundary tests. Give an example and a mistake that the idea helps you avoid.
- After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.
- This week you will practise:
- **Process** a list of records consistently
- Compute a total from named **field**s
- Preserve record meaning while tracing a loop
- Which key stores the occupied count in {"capacity": 8, "used": 3}?

My starting explanation

Your end goal

Trace reading structured data and explain the changed program before running it.

▼ Earlier wording, alternatives and picture transcripts

Your learning target

Remember before reading

Retrieve what you already know

This week’s end goal

MAIN LESSON · SECTION 2

Give every field and row policy a defined meaning

> A total is about the included records, not automatically the whole real-world collection.

Read the two-cupboard dataset

Two fictional cupboards are stored as a list of dictionaries. Each record uses the same keys and units. We want the total spare spaces. Treat the records as a **dataset**, and trace one record at a time before trusting a total.

State data and policy scope

Structured data are records organised into fields and rows. Before aggregating, define field meanings, units and missing-value policy. This lesson starts with supplied Python lists of dictionaries; it does not parse a file or validate arbitrary external input. For cupboard records, both capacity and used are required valid counts.

The pictured count-record exercise has a different explicit policy: include supplied numeric values, including zero, and report how many rows were measured.

Your first explanation

List the given facts and one fact that is not supplied.

MAIN LESSON · SECTION 3

Accumulate each record’s spare spaces

> Initialise total before processing the first row.

Complete capacity-minus-used record loop

```
rows = [{"capacity": 8, "used": 3},
        {"capacity": 6, "used": 6}]
total = 0
for row in rows:
    spare = row["capacity"] - row["used"]
    total = total + spare
print(total)
```

Trace each record’s contribution

Follow the evidence

STEP OR CASE	EVIDENCE	MEANING
First row	8 - 3 = 5	total becomes 5
Second row	6 - 6 = 0	total stays 5

Read each row from left to right and connect the evidence to the stated rule.

Explain empty-input and source limits

The loop variable row refers to each record in turn. We calculate spare from that record, then add it. total starts before the loop.

An empty list produces 0 because there are no updates; it does not tell us whether no cupboards exist or records were not collected.

Explain the deciding step

- Point to the **input**, the deciding step and the result in this week’s visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

▼ Earlier wording, alternatives and picture transcripts

Connect the picture and the explanation

Compare observed zero with an absent row field

> Skipping unavailable data changes the observed denominator and must be reported.

Predict the supported change

- For spare contributions 2, 4 and 1, trace total starting at 0.

> **Keep the procedure fixed. Explain what changed in the inputs and result.**

Predict the pictured variant

- Change: Add {"count":4} as a fourth row.

▼ **Check after predicting**

Prediction: Sum becomes 6, denominator 3, mean 2; coverage is 3 of 4.

Reason: Only the added recorded value contributes; the existing absent **field** remains skipped.

▼ **Earlier wording, alternatives and picture transcripts**

Compare a familiar case with a changed case

Change one thing in the pictured case

Trace included rows into a bounded observed statistic

> A real zero contributes a measured row even when it adds nothing to the sum.

Read the complete count-record policy

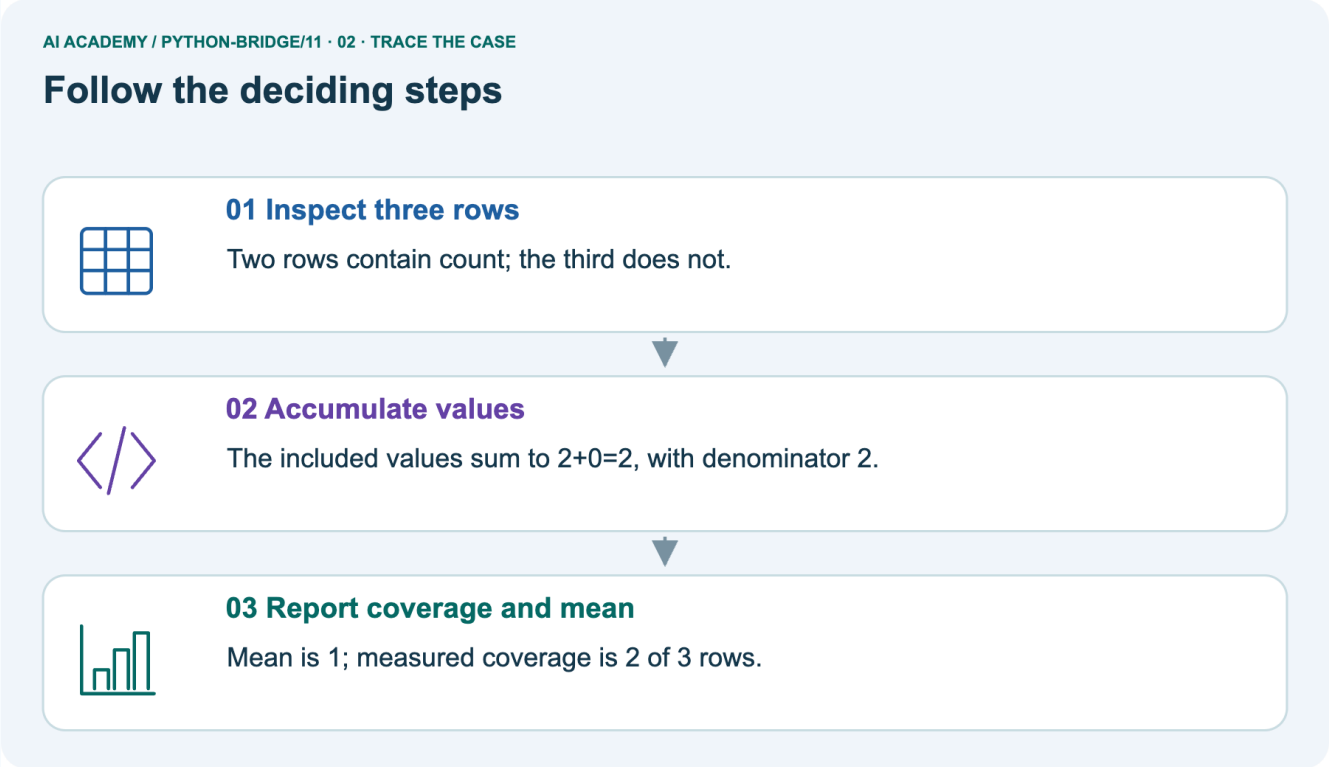
Read rows with a missing-**field** policy. A toy table is represented as a list of dictionaries; count only rows with a recorded count.

Python 3 records: [{"count":2},{"count":0},{}]. Policy: include existing numeric counts, including zero; skip an absent **field**.

The supplied rows contain count 2, count 0 and an absent field. append adds one value to the end of observed, retaining the real zero. sum(observed)adds included values; len(observed) is the mean denominator, while len(rows) is the number of offered records. With values 2 and 0, sum 2/denominator 2 gives mean 1.0 and coverage 2/3. None or an absent field is unavailable under this toy policy.

The program assumes supplied numeric values; it does not reject malformed types or verify measurements. An observed empty sum 0 does not prove missing population values are zero.

Trace included rows and the statistic



Follow the labelled picture with the matching explanation. Open the vector diagram

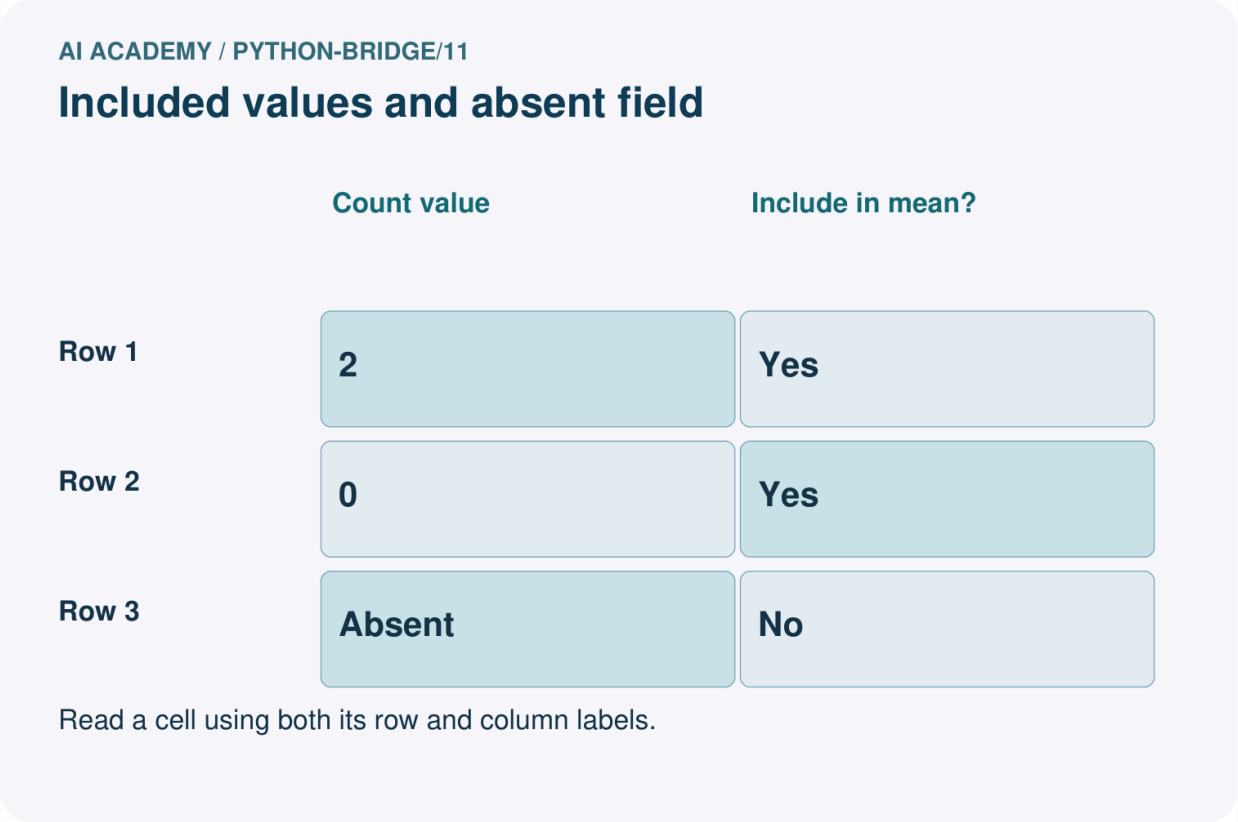
Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Inspect three rows	Two rows contain count; the third does not.
2. Accumulate values	The included values sum to 2+0=2, with denominator 2.
3. Report coverage and mean	Mean is 1; measured coverage is 2 of 3 rows.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Structured data need a row policy before aggregation. Treating missing as zero changes the statistic and can conceal incomplete coverage; real zero remains an **observation**.

Inspect the original include/absent graphic



Follow the labelled picture with the matching explanation. Open the vector diagram

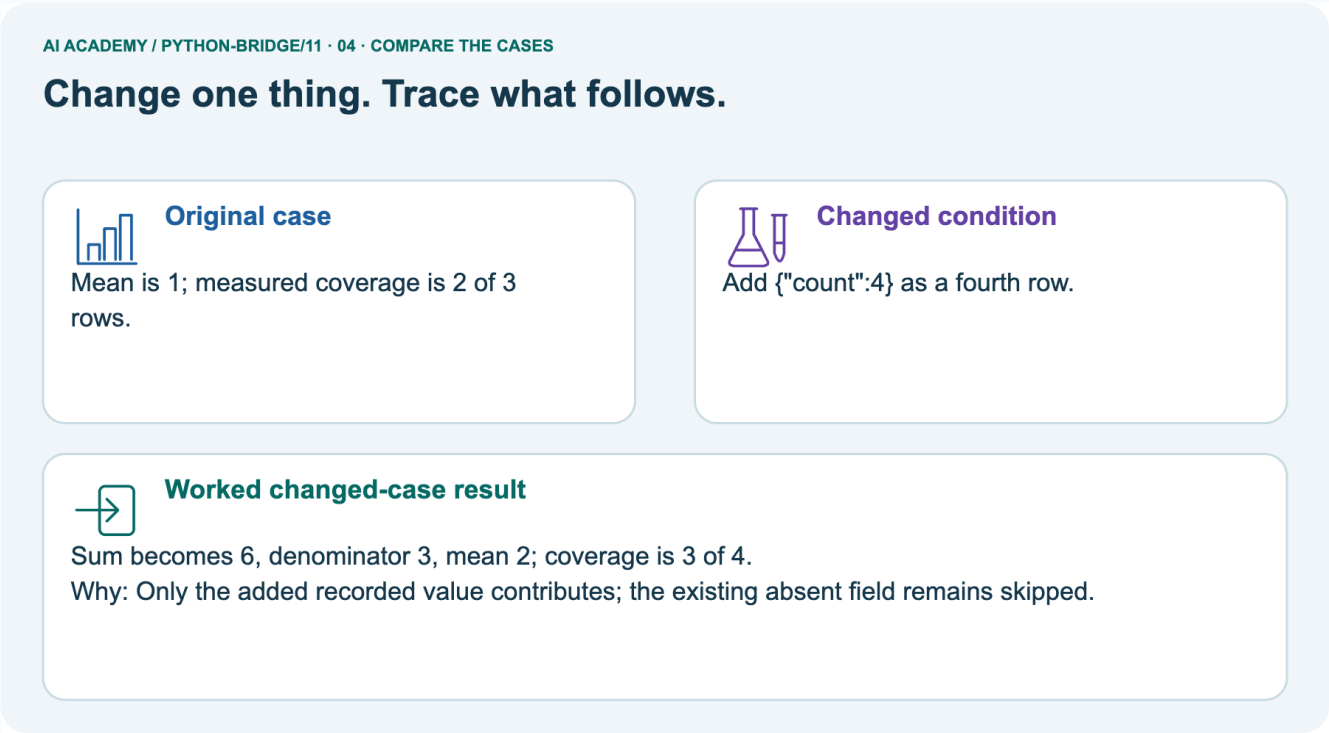
Complete total/denominator/coverage program

```
rows = [{"count": 2}, {"count": 0}, {}]
observed = []
for row in rows:
    value = row.get("count")
    if value is not None:
        observed.append(value)
observed_total = sum(observed)
if len(observed) > 0:
    mean = observed_total / len(observed)
else:
    mean = None
print(observed_total, len(observed), len(rows), mean)
```

Predict adding recorded count 4

- Change one thing: Add {"count":4} as a fourth row. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

State parsing, malformed-value and provenance limits

- This worked row policy is explicit; file parsing, malformed values and data provenance require additional checks.

- Read the labelled picture: Circle the included rows, calculate the sum and mean, and state coverage. Repeat after adding the fourth row with count four.

▼ Earlier wording, alternatives and picture transcripts

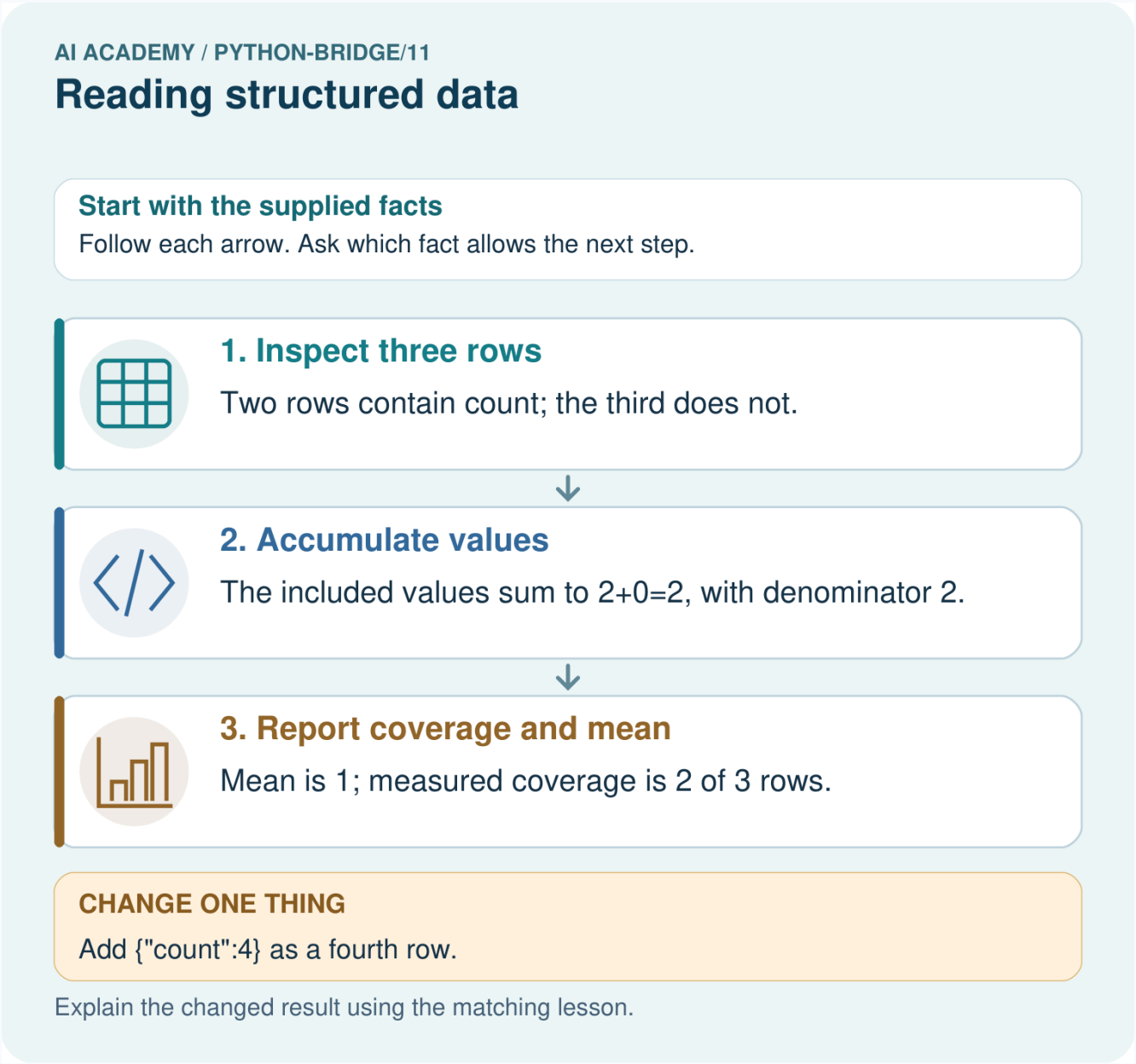
> **Add a third record with capacity 10 and used 4. Predict the total and list each contribution.**

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Reading structured data

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

MAIN LESSON · SECTION 7

Find a reset that discards earlier records

> The last contribution is not the aggregate of all contributions.

Explain the reset and reversed-order test

Putting total = 0 inside the loop discards earlier contributions. With the original rows it finishes at 0, the second contribution. Move the initialization before the loop and rerun both the original and reversed row order.

Repair the 2/4/1 contribution trace

- A loop resets total to 0 on each pass over contributions 2, 4, 1. What final total appears and how should it be fixed?

State one repair and the checks you would rerun.

▼ **Earlier wording, alternatives and picture transcripts**

Locate the earliest unsupported step

MAIN LESSON · SECTION 8

Practise an observed-record total with less help

> Keep the three W2 contributions distinct from the main two-cupboard rows.

Explain the record procedure

- In the worked **dataset**, state the number of records and the spare spaces in the second record.

> **Add a third record with capacity 10 and used 4. Predict the total and list each contribution.**

> **Explain your prediction to a partner before comparing results.**

Complete W2 running-total trace

Use workbook W2 for the connected example “Reading structured data”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Audit successful-run-as-valid-data reasoning

- Use the worked case for Reading structured data. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain included count rows and coverage

- Circle the included rows, calculate the sum and mean, and state coverage. Repeat after adding the fourth row with count four.

> **My explanation using a pictured step and the given facts**
>
> | |
> |---|
> | |
> | |
> | |
> | |

Reserve selected score records

This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ **Earlier wording, alternatives and picture transcripts**

Practise with less help: Trace all updates

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Explain the picture

MAIN LESSON · SECTION 9

Check six complete record and coverage cases

> A statistic needs its row policy and denominator next to the value.

All supplied records and expected boundaries

Complete fictional test	Expected aggregate and evidence limit
A · W2 records capacity/used(4,2),(7,3),(2,1)	Contributions 2,4,1; running totals 2,6,7
B · Main cupboard record loop on empty rows []	Procedural total 0; no evidence that actual cupboards are empty
C · Cupboard row lacks required used field	KeyError under required-field contract; do not invent unused capacity
D · Pictured count rows [2,0,missing]	Observed total 2, measured 2/3, mean 1.0
E · Add recorded count 4 to D	Observed total 6, measured 3/4, mean 2.0
F · Three empty count records	Observed sum 0, measured 0/3, mean None; actual missing values unknown

Complete W2 record fixture

```
rows = [{"capacity": 4, "used": 2},
        {"capacity": 7, "used": 3},
        {"capacity": 2, "used": 1}]
total = 0
for row in rows:
    spare = row["capacity"] - row["used"]
    total = total + spare
print(total)
```

Predict, run or trace the matching policy

1. Use the complete W2 fixture for A; it explicitly implements the given spare contributions without leaving capacity/used inputs unstated. Run or trace the main cupboard loop for B/C and the separate observed-count program for D/E/F. The policies differ: a missing required used field is an error, while unavailable count is skipped with coverage reported.

   Do not copy a skip policy onto a required field to make a failure disappear. Predict before running and preserve errors. Explain append, sum, included-row denominator and offered-row count. Empty observed sum 0 is an accumulator convention, not proof that unseen measurements are zero. Keep the selected score-record script for the first individual attempt.

Keep the evidence record

For spare contributions 2, 4 and 1, trace total starting at 0.

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or “not run”	
First mismatch and one explanation	
Boundary or missing input and expected response	

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ Reference · Original two-cupboard script

```
rows = [{"capacity": 8, "used": 3},
        {"capacity": 6, "used": 6}]
total = 0
for row in rows:
    spare = row["capacity"] - row["used"]
    total = total + spare
print(total)
```

▼ Earlier wording, alternatives and picture transcripts

Predict, investigate and preserve the evidence

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: report the selected score total, mean and coverage

> An unavailable mean must not become a fabricated zero.

Selected complete task and data policy

Trace the named **field**s and missing-value policy. Predict the **output**, state the observed total and mean denominator, and report coverage. What changes if all three rows are empty records?

```
rows = [{"score": 2}, {"score": 6}, {}]
values = []
for row in rows:
    value = row.get("score")
    if value is not None:
        values.append(value)
observed_total = sum(values)
```

```
if len(values) > 0:
    mean = observed_total / len(values)
else:
    mean = None
print(observed_total, len(values), len(rows), mean)
```

Use the selected score-field policy exactly: include supplied values when not None, preserve zero, and report the number of included scores as well as offered records. For the all-empty variant, distinguish the procedural observed sum from an unknown total of missing real scores. No file contents or measured outcomes beyond the stated rows are supplied.

Your response and help record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Use feedback afterwards

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ **Additional practice · Earlier record aggregation**

Write a loop for records (capacity 9, used 2), (capacity 5, used 1), and (capacity 4, used 4). Predict the total. Reverse the order and test an empty list.

My independent procedure, result and limitation

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Preserve records and report measured coverage

Attempt this without hints

Use feedback to improve

MAIN LESSON · SECTION 11

Keep each dataset’s policy in the portfolio

> A total without coverage can hide unrecorded values.

This week’s contribution

Add a labelled reading structured data **evidence** sheet to Python readiness portfolio. Show how you can **process** a list of records consistently.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week’s record; a matching answer without a trace is insufficient.

Project checkpoint and evidence

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

For Python readiness before the Responsible Plant Classifier, retain a trace of the values, the predicted **output**s and the changed-case explanation. Use actual Python 3 results to check the **prediction** after reasoning first.

▼ **Earlier wording, alternatives and picture transcripts**

Evidence for your project

MAIN LESSON · SECTION 16

Finish: name the included rows, denominator and coverage

> Observed aggregation is not knowledge of every missing value.

Explain without notes

- Close your notes. Explain this goal in your own words: **Process** a list of records consistently

My deciding step, evidence and one remaining question

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · The words used in this procedure**

REFERENCE · SECTION 5

Reference · The words used in this procedure

> Explain each term using a particular code line.

Vocabulary and response

Word	Meaning
Dataset	A collection of records
Row	One record in a table
Field	One named part of a record
Accumulator	A variable updated to keep a running result

Use two terms to explain a step in this week's example.

▼ Extra challenge · Explain, apply and challenge

REFERENCE · SECTION 10

Extra challenge · Explain, apply and challenge

> Choose the next useful task after supported practice.

Choose a challenge

- In the worked **dataset**, state the number of records and the spare spaces in the second record.
- A loop resets total to 0 on each pass over contributions 2, 4, 1. What final total appears and how should it be fixed?
- An empty **dataset** produces total 0. Can you conclude that every cupboard is full?

> **Explain which evidence supports your conclusion and which claim remains unproved.**

▼ Earlier wording, alternatives and picture transcripts

Explain → apply → challenge

▼ Reference · Inspect your evidence

REFERENCE · SECTION 12

Reference · Inspect your evidence

> Use the record after practice rather than revealing the independent answer.

Self-check and explanation

- This loop assumes every record has the required integer **field**s. A missing used **field** causes KeyError. Do not silently treat a missing count as zero; that would invent unused capacity.

- An empty **dataset** produces total 0. Can you conclude that every cupboard is full?

An empty dataset produces total 0. Can you conclude that every cupboard is full?

My limited claim and a useful next check

▼ Extra practice · A complete trace

REFERENCE · SECTION 13

Extra practice · A complete trace

> Predict before opening the supplied trace.

Explain the input and procedure

- Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

▼ Check the complete trace

Follow the evidence

STEP OR CASE	EVIDENCE	MEANING
First row	8 - 3 = 5	total becomes 5
Second row	6 - 6 = 0	total stays 5

Read each row from left to right and connect the evidence to the stated rule.

▼ Earlier wording, alternatives and picture transcripts

Complete evidence record

▼ Reference · Earlier route checklist

REFERENCE · SECTION 15

Reference · Earlier route checklist

> Keep one main lesson route and use this as a reminder.

Checklist and first prediction

By the end, I can	How I will show it
Process a list of records consistently	Explain the deciding step, show a trace or test, and state one limit.

By the end, I can	How I will show it
Compute a total from named fields	Explain the deciding step, show a trace or test, and state one limit.
Preserve record meaning while tracing a loop	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ **Earlier wording, alternatives and picture transcripts**

Start with this question: Read rows with a missing-**field** policy. A toy table is represented as a list of dictionaries; count only rows with a recorded count.

▼ **Extra practice · Predict, check and repair**

REFERENCE · SECTION 17

Extra practice · Predict, check and repair

> Use only after an attempt; keep the reserved assessment separate.

First question and hint

- Read rows with a missing-**field** policy. A toy table is represented as a list of dictionaries; count only rows with a recorded count.
- Which labelled step matches this detail? Mean is 1; measured coverage is 2 of 3 rows.
- 1. Inspect three rows

- 2. Report coverage and mean
- 3. Accumulate values
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**
>
> | |
> |---|
> | |
> | |
> | |
> | |

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Report coverage and mean
Mean is 1; measured coverage is 2 of 3 rows. Structured data need a row policy before aggregation.

Treating missing as zero changes the statistic and can conceal incomplete coverage; real zero remains an **observation**.

Changed-case question and hint

- Changed-case practice: Add {"count":4} as a fourth row. Which conclusion is supported by the supplied facts?
- 1. We can decide the result without examining the changed **input** or the procedure.
- 2. This one example guarantees correct results for every future **input**.
- 3. Sum becomes 6, denominator 3, mean 2; coverage is 3 of 4.

- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check the changed explanation**

After your attempt, compare with the supported explanation: Sum becomes 6, denominator 3, mean 2; coverage is 3 of 4. Only the added recorded value contributes; the existing absent **field** remains skipped.

Repair and keep the record

Review Dictionaries and records (python-bridge/10). Goal: Read and update named dictionary **field**s. Short practice: Predict all three original **output**s.

Explain the difference between count=0 and missing price, then predict the lookup after price=0 is added.. Return to this week and explain what you changed.

Review Functions and arguments (python-bridge/8). Goal: Distinguish a function definition from a function call. Short practice: Match every argument to a parameter for the two calls.

Why do **output**s differ, and what prints when both calls are removed?. Return to this week and explain what you changed.

Review Branching and boundary tests (python-bridge/5). Goal: Trace the selected branch in an if elif else chain. Short practice: Trace -2, 4 and 5 through the branches.

Explain why the negative check comes first and what label -2 gets if it is removed.. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

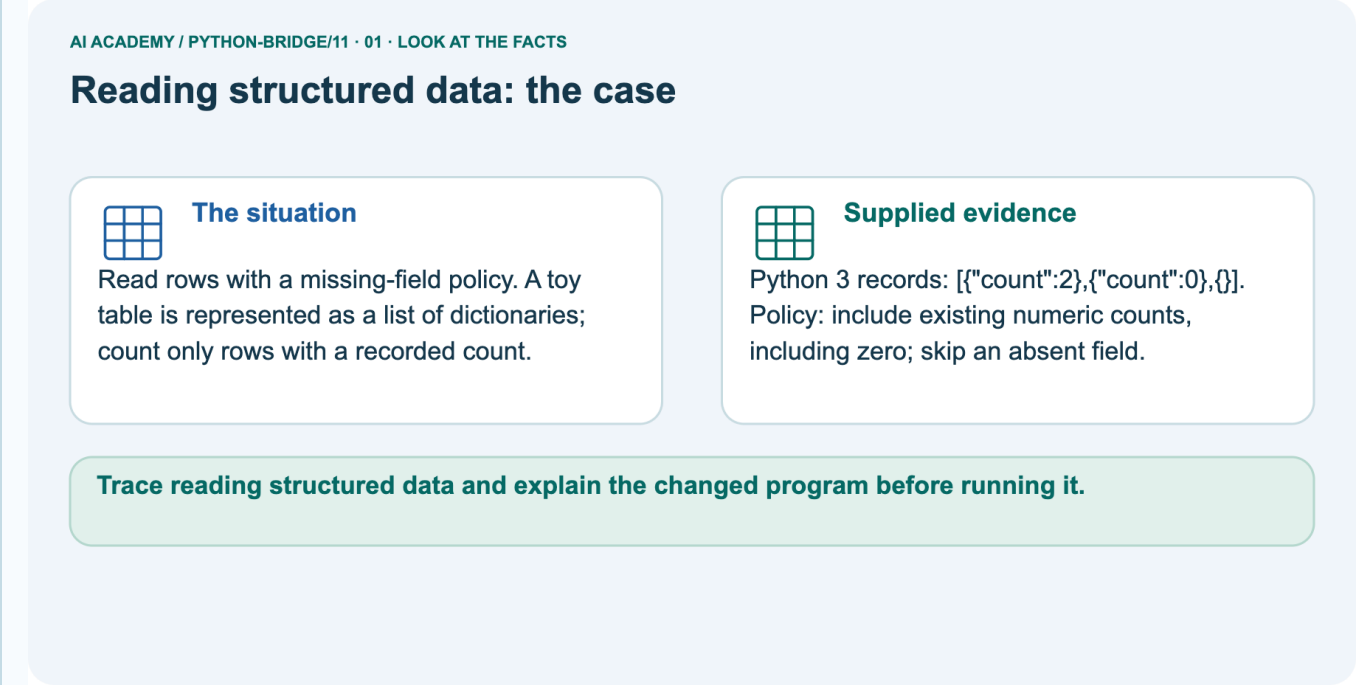
▼ **Reference · The six-picture journey**

REFERENCE · SECTION 18

Reference · The six-picture journey

> Choose the picture needed for the present question.

01. Reading structured data: the case



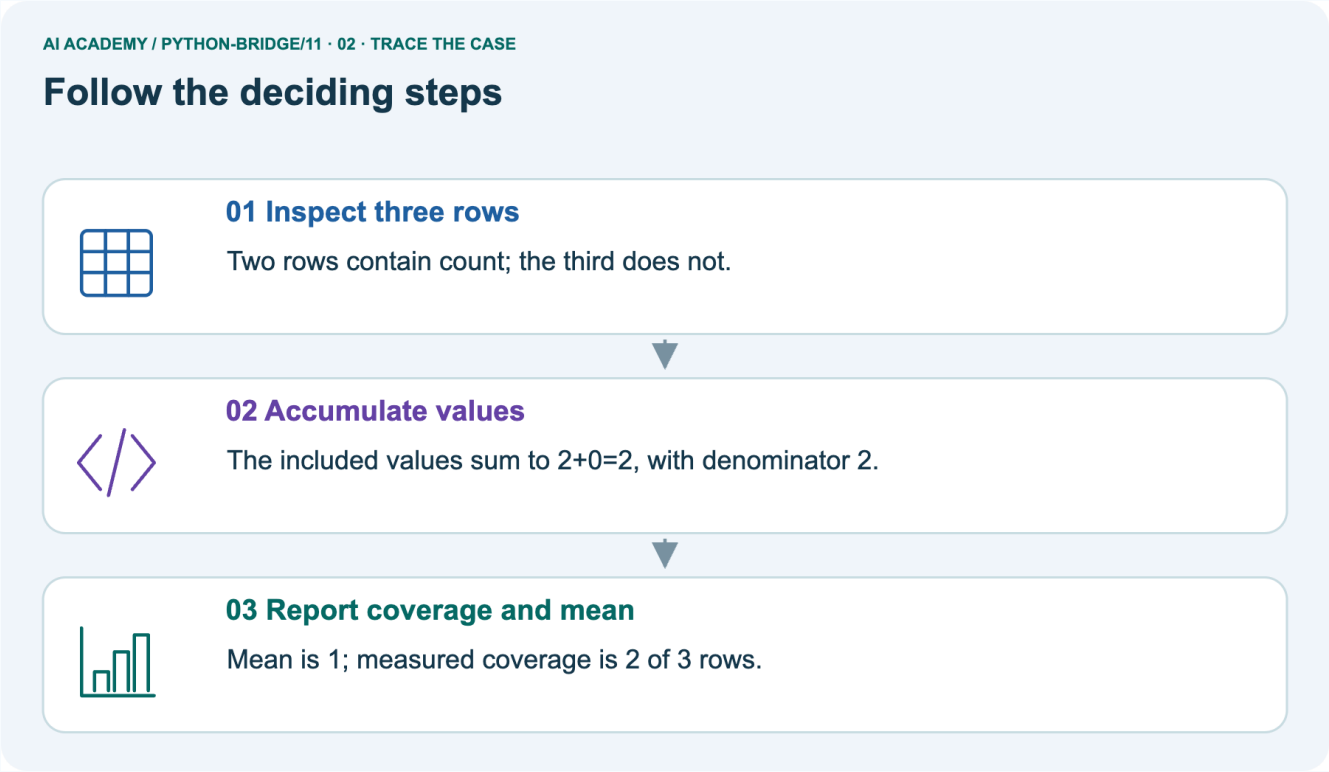
Follow the labelled picture with the matching explanation. Open the vector diagram

Read rows with a missing-**field** policy. A toy table is represented as a list of dictionaries; count only rows with a recorded count. Supplied facts: Python 3 records: [{"count":2}, {"count":0},{}].

Policy: include existing numeric counts, including zero; skip an absent **field**.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

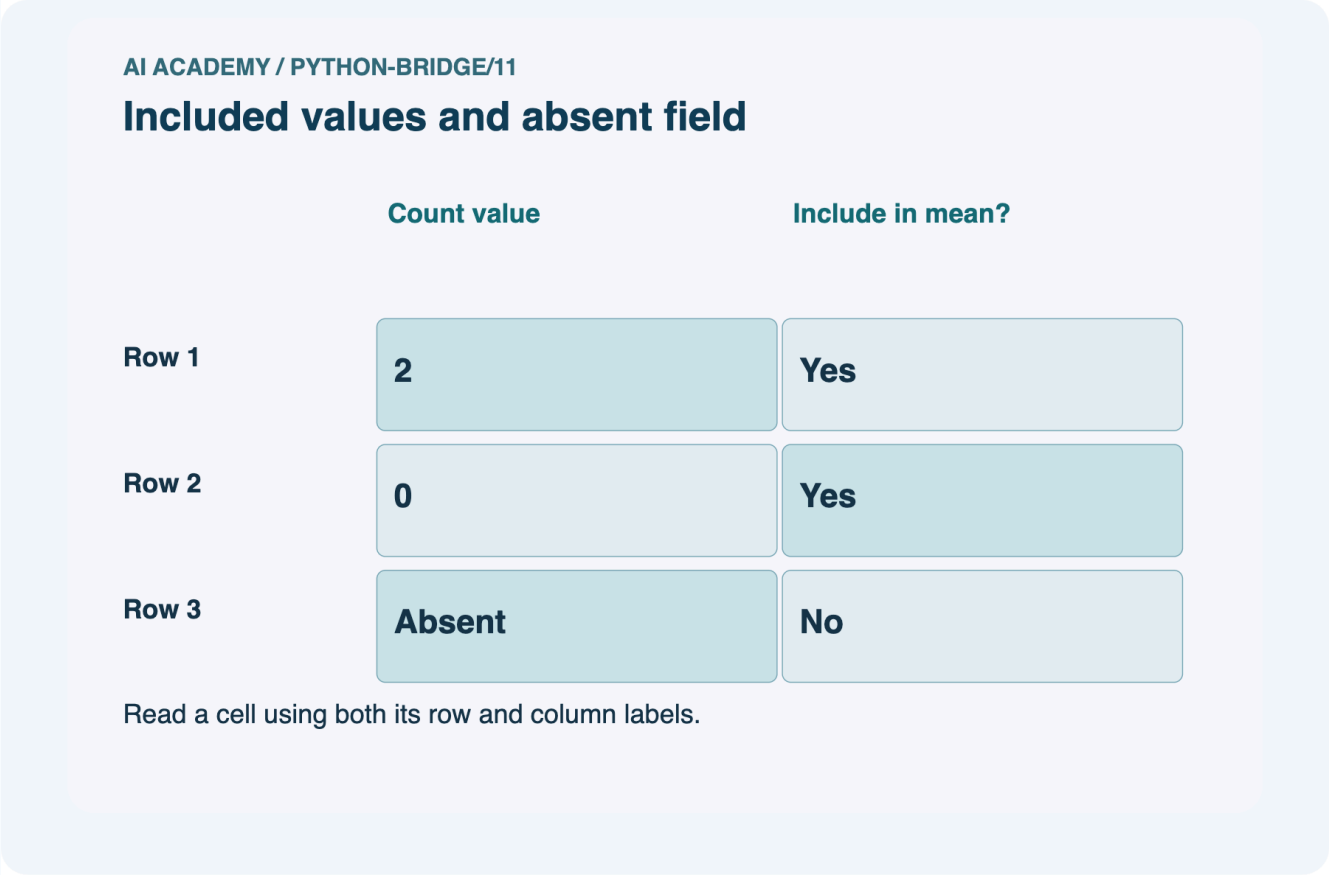


Follow the labelled picture with the matching explanation. Open the vector diagram

Inspect three rows: Two rows contain count; the third does not. → Accumulate values: The included values sum to 2+0=2, with denominator 2. → Report coverage and mean: Mean is 1; measured coverage is 2 of 3 rows.

Point to each numbered stage and explain which recorded **input** it uses.

03. Included values and absent field

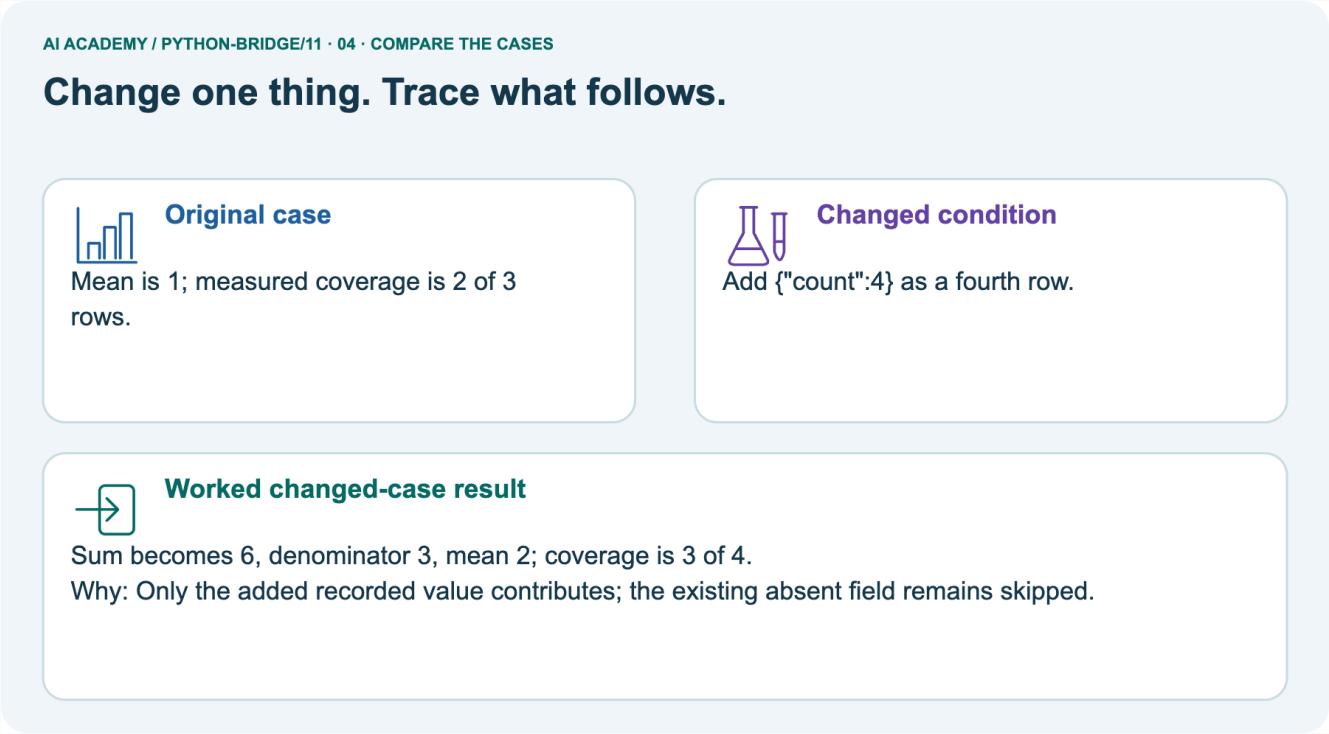


Follow the labelled picture with the matching explanation. Open the vector diagram

Included values and absent **field**. Row 1: Count value = 2, Include in mean? = Yes. Row 2: Count value = 0, Include in mean? = Yes. Row 3: Count value = Absent, Include in mean? = No.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



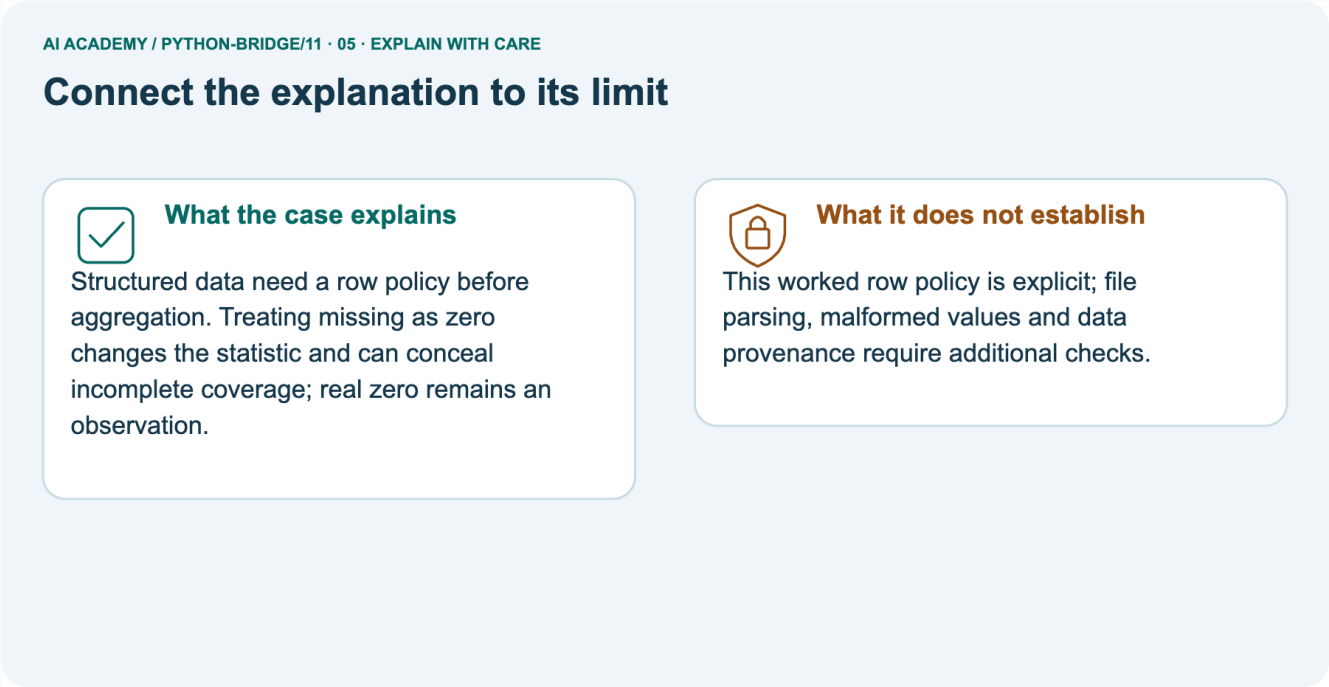
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Mean is 1; measured coverage is 2 of 3 rows. Changed condition: Add {"count":4} as a fourth row. Expected result: Sum becomes 6, denominator 3, mean 2; coverage is 3 of 4.

Reason: Only the added recorded value contributes; the existing absent **field** remains skipped.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



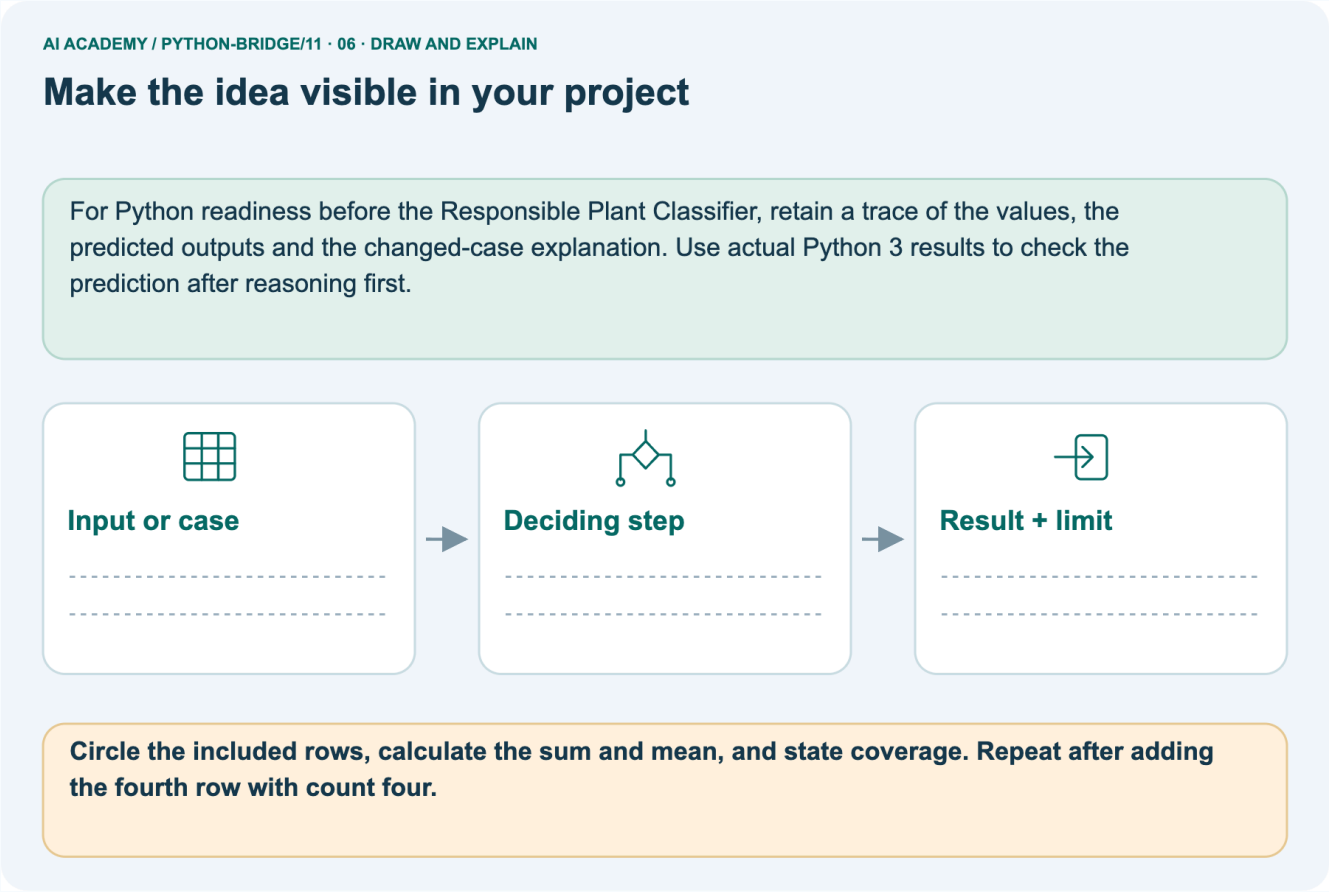
Follow the labelled picture with the matching explanation. Open the vector diagram

Structured data need a row policy before aggregation. Treating missing as zero changes the statistic and can conceal incomplete coverage; real zero remains an **observation**.

Limit: This worked row policy is explicit; file parsing, malformed values and data provenance require additional checks.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For Python readiness before the Responsible Plant Classifier, retain a trace of the values, the predicted **output**s and the changed-case explanation. Use actual Python 3 results to check the **prediction** after reasoning first.

Draw your **input**, deciding step, result and limit. Supported question: Circle the included rows, calculate the sum and mean, and state coverage. Repeat after adding the fourth row with count four.

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Reading structured data: the case

02. Follow the deciding steps

03. Included values and absent field

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.